

Experiment No: 5

Aim: To study the parallel operation of two single-phase transformer.

Theory:

When electric power is supplied to a locality, city or an area, a single transformer, capable of handling the required power demand, is installed. In some cases, it may be preferable to install two or more transformers in parallel, instead of one large unit. Though two or more transformers may be expensive than one large unit, yet this scheme possesses certain advantages described below:

- (1) With two or more transformers, the power system becomes more reliable. For instance if one transformer develops fault, it can be removed and the other transformers can maintain the flow of power, though at a reduced level.
- (2) Transformers can be switched off or on, depending upon the power demand. In this manner, the transformer losses decrease and the system becomes more economical and efficient in operation.
- (3) The cost of a standby (or spare) unit is much less when two or more transformers are installed.

In any case, with the passage of time, electric power demand may become more than the rated kVA capacity of the already existing transformer or transformers. Under such circumstances, the need for extra transformer arises. Since the supply voltage has to remain constant, the extra unit must be connected in parallel.

Note that the parallel operation of transformers requires that their primary windings as well as their secondary windings, are connected in parallel.

The various conditions which must be fulfilled for the satisfactory parallel operation of two or more single-phase transformers, are as follows :

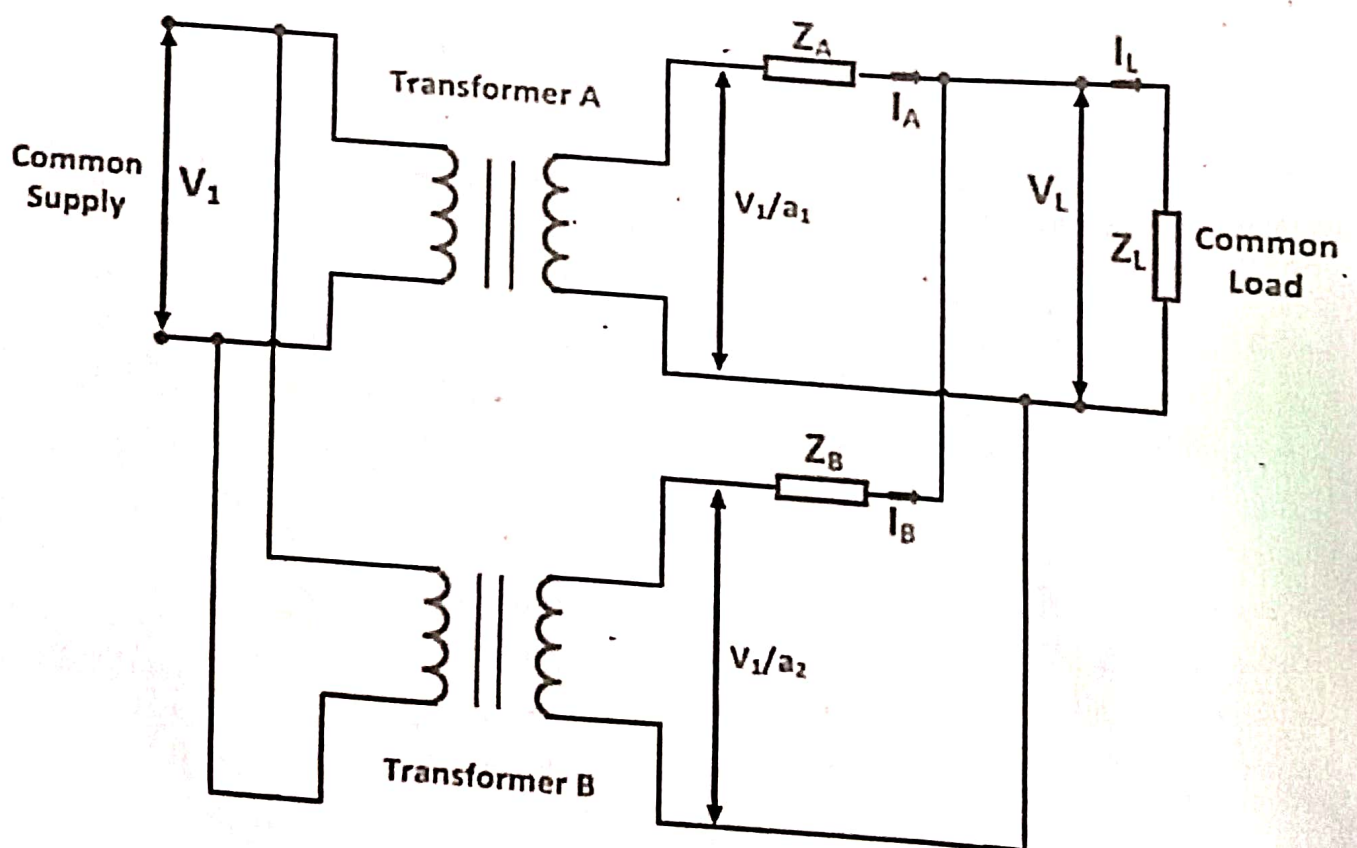
- (A) The transformers must have the same voltage ratios, i.e. with the primaries connected to the same voltage source, the secondary voltages of all transformers should equal in magnitude.
- (B) The equivalent leakage impedances in ohms should be inversely proportional to their respective kVA ratings. In other words, the per unit leakage impedances of the transformers based on their own kVA ratings must be equal.

(C) The ratio of equivalent leakage reactance to equivalent resistance, i.e. x/r should be same for all the transformers.

(D) The transformers must be connected properly, so far as their polarities are concerned.

Out of the conditions listed above, condition (D) must be strictly fulfilled. If the secondary terminals are connected with wrong polarities, large circulating currents will flow and the transformers may get damaged. Condition (A) should be satisfied as accurately as possible; since different secondary voltages would give rise to undesired circulating currents. For conditions (B) and (C), some deviation is permissible. Thus the fulfilment of condition (D) is essential, whereas the fulfilment of other conditions is desirable.

The circuit diagram of two single-phase transformer A and B connected in parallel are shown below:



Let,

a_1 is the turn ratio of the transformer A

a_2 is the turn ratio of transformer B

Z_A is the equivalent impedance of the transformer A referred to secondary

Z_B is the equivalent impedance of the transformer B referred to secondary

Z_L is the load impedance across the secondary

I_A is the current supplied to the load by the secondary of the transformer A

I_B is the current supplied to the load by the secondary of the transformer B

V_L is the secondary load voltage

I_L is the load current

Applying Kirchhoff's Current Law,

$$I_A + I_B = I_L \dots \dots \dots (1)$$

By Kirchhoff's Voltage Law,

$$V_L = \frac{V_1}{a_1} - I_A Z_A \dots \dots \dots (2) \text{ and}$$

$$V_L = \frac{V_1}{a_2} - I_B Z_B \dots \dots \dots (3)$$

Now putting the value of I_B from the equation (1) in equation (3) we will get,

$$V_L = \frac{V_1}{a_2} - (I_L - I_A) Z_B \dots \dots \dots (4)$$

Solving equations (2) and (4) we will get,

$$I_A = \frac{Z_B I_L}{Z_A + Z_B} + \frac{V_1 (a_2 - a_1)}{a_1 a_2 (Z_A + Z_B)} \dots \dots \dots (5)$$

$$I_B = \frac{Z_A I_L}{Z_A + Z_B} + \frac{V_1 (a_2 - a_1)}{a_1 a_2 (Z_A + Z_B)} \dots \dots \dots (6)$$

The current I_A and I_B have two components. The first component represents the transformers share of the load currents and the second component is a circulating current in the secondary windings of the single-phase transformer.

Equal Voltage Ratio

In order to eliminate circulating currents, the voltage ratios must be identical. That is $a_1 = a_2$

$$I_A = \frac{Z_B I_L}{Z_A + Z_B} \dots \dots (7)$$

$$I_B = \frac{Z_A I_L}{Z_A + Z_B} \dots \dots (8)$$

Under this condition,

$$\frac{I_A}{I_B} = \frac{Z_B}{Z_A} \dots \dots (9)$$

Equating equation (7) and (8) we will get:

From the above equation (9), it is clear that the transformer currents are inversely proportional to the transformer impedance. Thus, for the efficient parallel operation of the two single-phase transformers, the potential differences at full load across the transformer internal impedance should be equal.

This condition ensures that the load sharing between the two single-phase transformer is according to the rating of each transformer. If the per-unit equivalent impedance are not equal, then the transformer will not share the load in proportion to their kVA ratings. As a result, the overall rating of the transformer bank will be reduced. Equation (9) can also be written as

$$I_A Z_A = I_B Z_B \dots \dots (10)$$

The current in the equations (7) and (8) is changed into volt-amperes by multiplying the two equations by the common load voltage V_L .

Therefore, we know that,

The total load in volt-ampere (VA) is

$$S_L = V_L I_L$$

The volt-ampere of transformer A is

$$S_A = V_L I_A$$

Similarly, the volt-ampere of transformer B is

$$S_B = V_L I_B$$

Hence, the various equations will be written as shown below

$$S_A = \frac{Z_B}{Z_A + Z_B} S_L \dots\dots\dots(11)$$

$$S_B = \frac{Z_A}{Z_A + Z_B} S_L \dots\dots\dots(12)$$

Equating the equation (11) and (12) we will get

$$\frac{S_A}{S_B} = \frac{Z_B}{Z_A} \dots\dots\dots(13)$$

Equation (13) tells that the volt-ampere load on each single-phase transformer is inversely proportional to its impedance.

Hence, to share the load in proportion to their ratings, the transformers should have the impedance which is inversely proportional to their ratings.